

📖 Extra — Creating Dictionaries: Advanced

Topic: Lists of Dictionaries + Nesting · **Course:** Extra Worksheet · **Time:** about 40 minutes

Real programs hold many records at once — a whole team, not just one hero. You put a **dictionary inside a list**, or even a **dictionary inside a dictionary**. On paper you will read, predict, fix, and explain. Then you will write and run your own code in the IDE.

Keep these words handy: **list of dictionaries**, **nested**, **.get()**, **KeyError**, **update**.

1 · Predict 🧠

Read each snippet. Write what you think it will print.

```
team = [
    {"name": "Aria", "hp": 100},
    {"name": "Leo", "hp": 80},
]
print(team[1]["name"])
```

Two steps: first `[1]`, then `["name"]`. What prints?


```
player = {
    "name": "Mia",
    "stats": {"hp": 100, "attack": 30},
}
print(player["stats"]["attack"])
```

This is a dictionary inside a dictionary. What prints?


```
player = {"name": "Mia"}  
print(player.get("score", 0))
```

There is no **"score"** key, but this does not crash. What prints, and why?

2 · Spot the Bug 🐛

Each snippet is broken. Read what it should do, then rewrite it so it works. Explain why the original was wrong.

Bug A — This should print **every** member's name. Right now it prints the same one three times.

```
team = [  
    {"name": "Aria"},  
    {"name": "Leo"},  
    {"name": "Mia"},  
]  
  
for member in team:  
    print(team[0]["name"])
```

Hint: use the loop variable `member`, not `team[0]`.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — This should read the nested attack value. It crashes because it reaches for `"attack"` at the wrong level.

```
player = {"name": "Mia", "stats": {"hp": 100, "attack": 30}}  
print(player["attack"])
```

Hint: `"attack"` lives inside `"stats"`, not directly inside `player`.

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Explain the Code

```
roster = [  
    {"name": "Aria", "role": "healer"},  
    {"name": "Leo", "role": "tank"},  
]  
  
new_hero = {}  
new_hero["name"] = "Mia"  
new_hero["role"] = "mage"  
roster.append(new_hero)  
  
for hero in roster:  
    print(f"{hero['name']} - {hero['role']}")
```

What is **roster** — and what is each item inside it?

How is **new_hero** built before it joins the roster?

What does the loop print, and why does it now include Mia?

4 · Code It

Open the IDE at **app.english-coding.co.uk** and write your own program.

Build a **list of dictionaries** called `shop`, where each item is a dictionary with a `"name"` key and a `"price"` key. Put three items in it.

Then **loop over** the shop and print each item's name and price on its own line.

Run it, then write your code and what it printed.

Submit 

Send this worksheet + a screenshot of your working code to teacher on KakaoTalk.